



histopathology cell injury

3rd class Lecture II

Assistant Prof. Dr. Rasha A. AL Safi

Histopathologist

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Objectives

By the end of this lecture, students will be able to

1. Analyze the types of cellular adaptation—hypertrophy, hyperplasia, atrophy, and metaplasia—by comparing their causes, mechanisms, and examples.
2. Evaluate the clinical significance of intracellular accumulations (e.g., lipofuscin, hemosiderin, bilirubin) in liver and cardiac pathology contexts.
3. Construct a concept map summarizing subcellular responses to injury (autophagy, mitochondrial changes, cytoskeletal abnormalities) and their pathological relevance.
4. Recall the main types of pathologic calcification (dystrophic and metastatic), their causes, and affected tissues, with at least two gross or microscopic features.

SUBCELLULAR RESPONSES TO INJURY

I. Autophagy:

- Autophagy is an adaptation to **nutrient deprivation**.
- Refers to **lysosomal digestion** of the cell's own components.
- The organelles are sequestered from the cytoplasm in an **autophagic vacuole**.
- The vacuole fuses with lysosomes to form **phagolysosome**, & the cellular components are digested by lysosomal enzymes.

II. Induction of smooth endoplasmic reticulum

III. Mitochondrial Alterations:

- An increase in their number in cellular hypertrophy.
- A decrease in number during cellular atrophy.
- Extremely large and abnormal shapes (megamitochondria), e.g. in hepatocytes in association with nutritional deficiencies and alcoholic liver disease.

III. Cytoskeletal Abnormalities:

CELLULAR ADAPTATIONS TO STRESS

Adaptations: are reversible changes in the **number, size, phenotype, metabolic activity, or functions** of cells in response to changes in their environment.

- I. **Physiologic adaptations:** usually represent responses of cells to **normal stimulation** by:
 1. **Hormones** or
 2. Endogenous **chemical mediators**
(e.g., the hormone-induced enlargement of the **breast** and **uterus** during pregnancy).
- II. **Pathologic adaptations** are responses to:
 - **Stress**
 - Allow cells to modulate their structure and function and thus **escape injury**.

HYPERTROPHY

Hypertrophy is an **increase in the size** of cells resulting in increase in the size of the organ.

In **pure hypertrophy**: there are **no new cells**, just bigger cells containing increased amounts of structural proteins and organelles.

- **Hypertrophy** occurs when cells have a **limited capacity to divide**.
- **Hypertrophy** can be **physiologic** or **pathologic**.

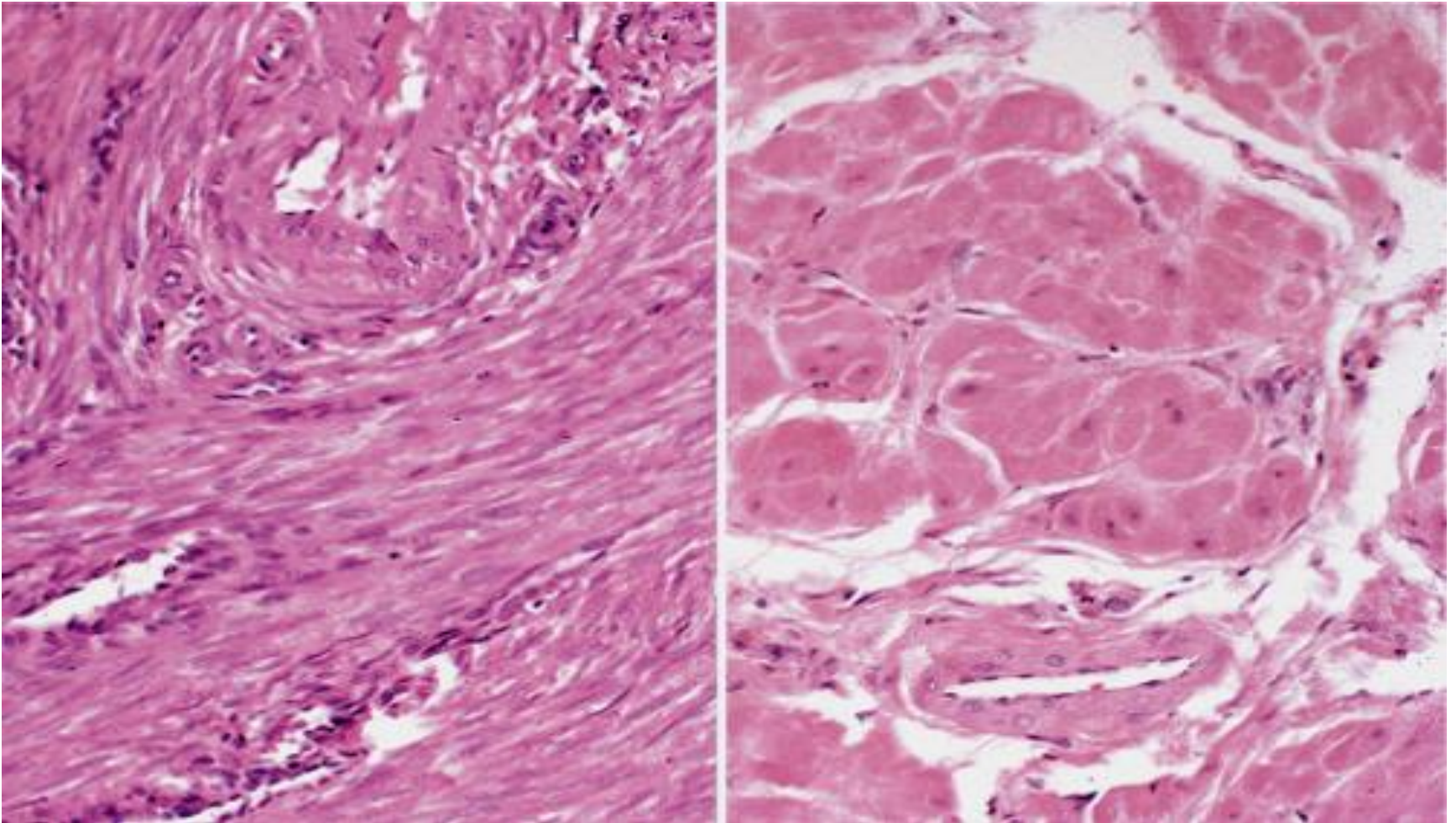
■ **Physiological Hypertrophy is caused:**

1. Increased **functional demand**
2. **GF**
3. **Hormonal** stimulation.

- E.g.: The massive **physiologic enlargement of the uterus** during pregnancy due to **estrogen** stimulated smooth muscle **hypertrophy** and smooth muscle **hyperplasia**.



Normal Vs hypertrophied uterine smooth muscle cells



Lt. small spindle-shaped uterine smooth muscle cells from a normal uterus.

Compare this with (Rt) large, plump hypertrophied smooth muscle cells from a gravid uterus .

An example of

Pathological Hypertrophy:

cardiac enlargement

hypertension

aortic valve disease

- A **limit** is reached **beyond** which the enlargement of muscle mass **can no longer compensate** for the increased burden.
- When this happens in the **heart**, “**degenerative**” **changes** occur in the myocardial fibers.
- The net **result** of these changes is **ventricular dilation** and **HF**.

Lt. ventricular hypertrophy



This is cardiac hypertrophy.

The **number of myocardial fibers never increases**, but their size can increase in response to an increased workload, leading to the marked thickening of the left ventricle in this patient with hypertension.

Note: normal Lt. ventricular wall thickness is 1.2 cm. to 1.5 cm

HYPERPLASIA

- Hyperplasia is an **increase in the number** of cells, takes place if the tissue contains cell populations capable of replication;
- it may occur concurrently with hypertrophy and often in response to the same stimuli.
- Hyperplasia can be physiologic or pathologic.
- In both situations, cellular proliferation is stimulated by GF that are produced by a variety of cell types.

I. Physiologic hyperplasia :

- (1) **Hormonal hyperplasia**: the proliferation of the glandular epithelium of the female breast at puberty and during pregnancy,
- (2) **Compensatory hyperplasia**: in which residual tissue grows after removal or loss of part of an organ.

Example (CH)

- When part of a liver is resected, **mitotic activity** in the remaining cells begins as early as 12 hours later,
- Eventually restoring the liver to its **normal weight**.
- The **stimuli** for hyperplasia in this setting are **GF** produced by un-injured hepatocytes.
- After restoration of the liver mass, cell proliferation is “**turned off**”.

II. Pathologic hyperplasia: Are caused by excessive **hormonal** or **GF** stimulation. E.g.:

1. Hyperplasia of connective tissue cells in **wound healing**.
2. Hyperplasia that is associated with certain **viral infections**; for example, human papillomaviruses (**HPV**) cause **skin warts**.
3. **Benign prostatic hyperplasia**: induced in responses to hormonal stimulation by androgens.
4. After a normal menstrual period there is a burst of **uterine epithelial proliferation** that is normally tightly regulated by the **stimulatory** effects of **pituitary hormones** and **ovarian estrogen** and the **inhibitory** effects of **progesterone**.

A disturbance in this balance leading to increased estrogenic stimulation causes **endometrial hyperplasia**.

An important point is that:

- the **hyperplastic process** should be remained **controlled**.
- It is this **responsiveness** to normal regulatory control mechanisms that distinguishes:
 1. Pathologic hyperplasia from
 2. Cancer.

Endometrial hyperplasia



The prominent folds of endometrium in this uterus are an example of hyperplasia.

The hyperplasia involves both endometrial glands and stroma.

ATROPHY

- **Atrophy: Shrinkage** in the **size** of the cell by the loss of cell substance.
- When a sufficient number of cells are involved, the entire tissue or organ diminishes in size, becoming atrophic.

Causes:

1. A decreased workload (e.g., immobilization of a limb),
2. loss of innervation.
3. Diminished blood supply.
4. Inadequate nutrition.
5. loss of endocrine stimulation.
6. Aging (senile atrophy).



Atrophy as seen in the brain.

A, Normal brain of a young adult.

B, Atrophy of the brain in an 82-year-old man with atherosclerotic disease.

- Atrophy of the brain is due to aging and reduced blood supply. Note that loss of brain substance narrows the gyri and widens the sulci.

METAPLASIA

- **Metaplasia** is a reversible change in which one adult cell type is replaced by another adult cell type.
- In this type of cellular adaptation, a **cell type** sensitive to a particular stress is **replaced** by another cell type **better able to withstand the adverse environment**.

- **Epithelial metaplasia** is exemplified by:

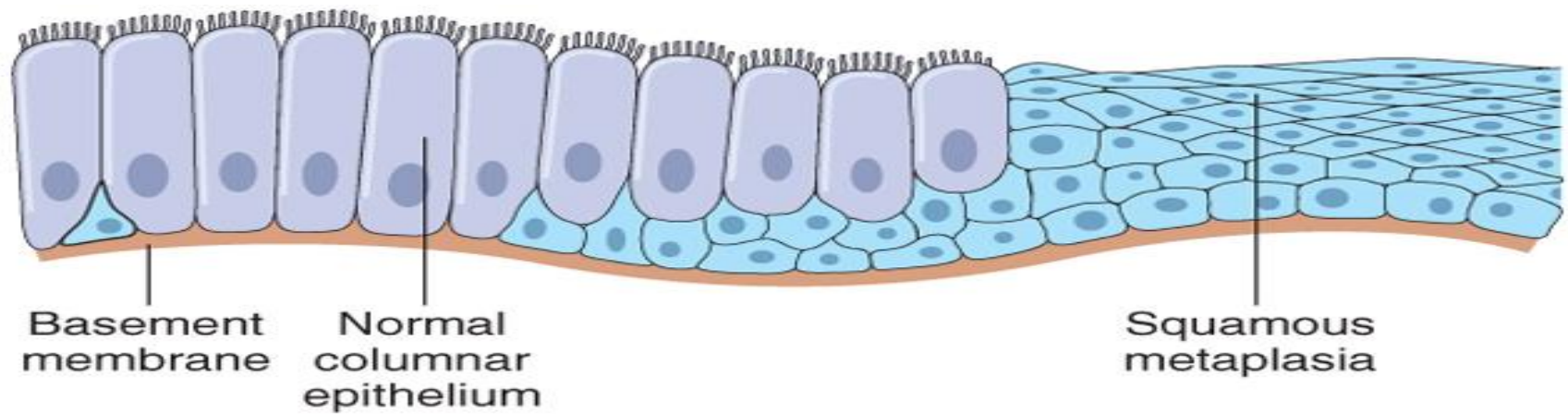
1. **Habitual cigarette smokers:**

- The respiratory epithelium of those may show a **Squamous changes** (metaplasia).
- The **normal** ciliated columnar epithelial cells of the trachea and bronchi are focally or widely **replaced by** stratified squamous epithelial cells with survival advantages.
- The important protective mechanisms are **lost**, (mucus secretion and ciliary clearance of particulate matter).
- If persistent, may predispose to **malignant transformation** of the epithelium.
- Cigarette smoking initially causes **squamous metaplasia**, and **cancers** arise later in **some** of these altered foci.

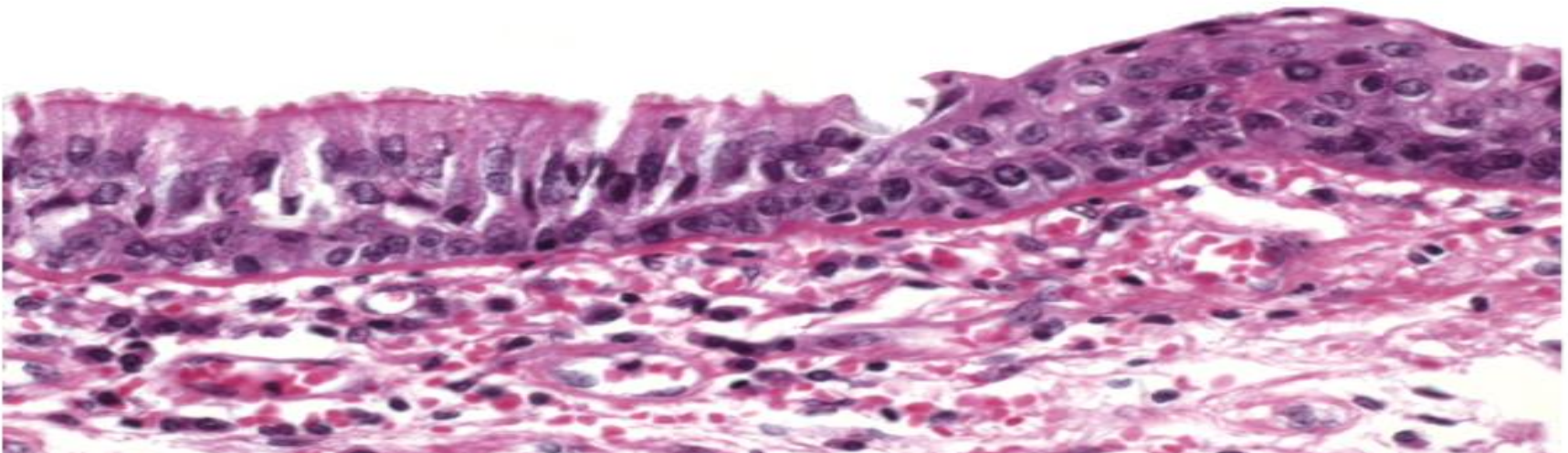
1. **Chronic gastric reflux:**

- The normal stratified squamous epithelium of the lower esophagus may undergo metaplastic transformation to gastric or intestinal-type columnar epithelium.

Epithelial Metaplasia



A



Metaplasia of normal columnar (left) to squamous epithelium (right) in a bronchus, shown (above) schematically and (down) histologically.

Intracellular accumulations:

- Cells may accumulate **abnormal amounts** of various substances; these may be harmless or associated with injury.
- The locations of these substances are either **cytoplasmic** within organelles (typically lysosomes), or in the **nucleus**.

A. Fatty Change (Steatosis)

- This refers to any **abnormal** accumulation of **triglycerides** within parenchymal cells.
- It is most often seen in the **liver**, but it may also occur in the **heart**.

Causes of fatty change include:

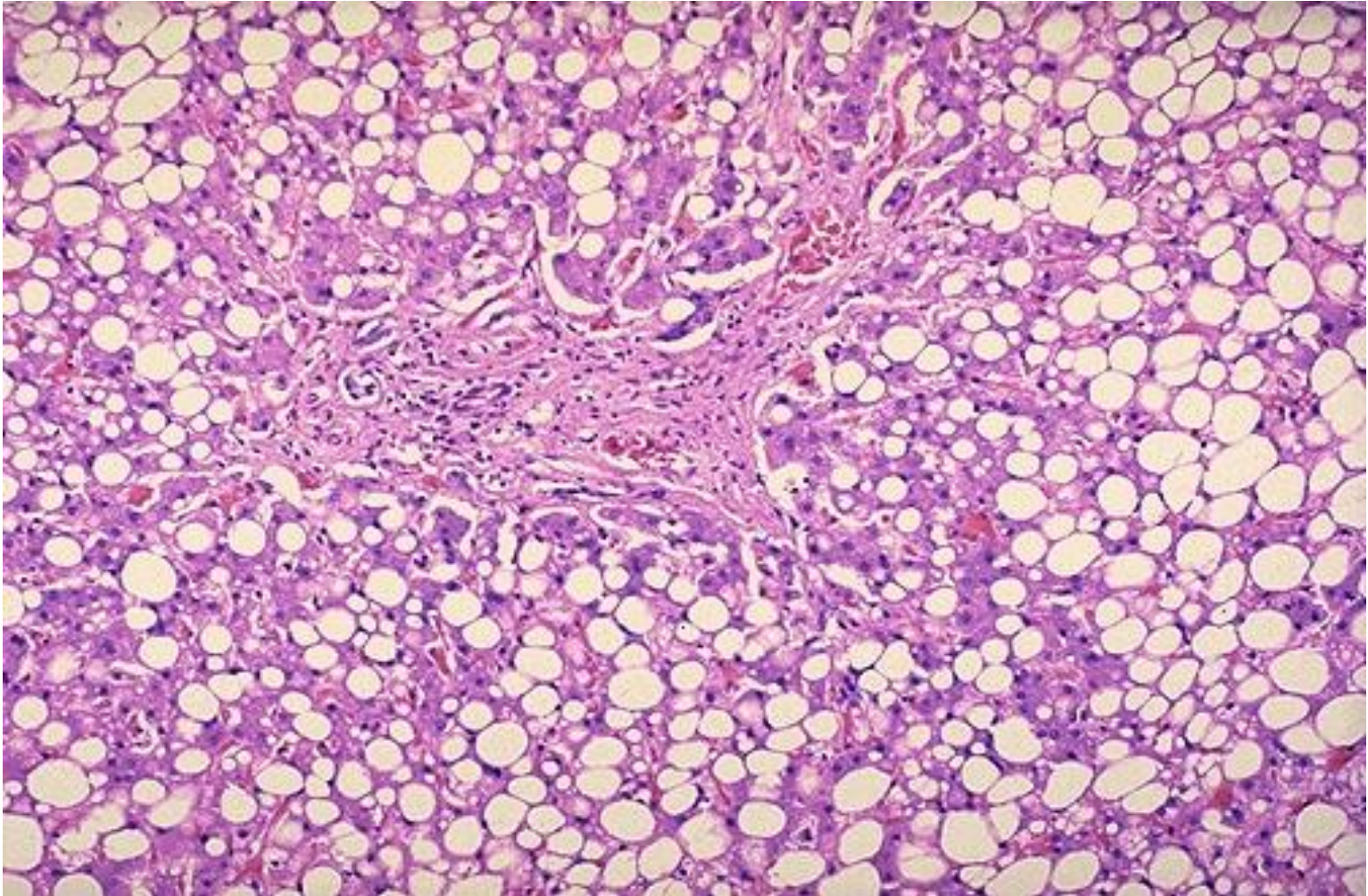
1. Alcohol.
 2. Diabetes mellitus.
 3. Obesity.
 4. Protein malnutrition.
 5. Anoxia.
- The significance of fatty change depends on **the cause** and **severity of the accumulation**.

Severe fatty change liver



In the liver mild fatty change shows no gross changes.
with progressive accumulation, the organ enlarges and become increasingly yellow,
soft and greasy to touch.

Fatty change liver



In the early stages: small fat vacuoles around the nucleus.

with progression these coalesce into a large clear space that displaces the nucleus to the periphery (macrovesicular steatosis).

B. Pigments

- Pigments are colored substances, exogenous or endogenous.

I. Exogenous pigments:

- The most common of these is **carbon** (an example is coal dust).
- When inhaled,
 - it is phagocytosed by **alveolar macrophages**.
 - transported through lymphatic channels to the regional tracheobronchial lymph nodes.
 - Aggregates of the pigment **blacken** the draining lymph nodes and pulmonary parenchyma (**anthracosis**).
 - Heavy accumulations may induce fibroblastic reaction that can result in a serious lung disease called **coal workers' pneumoconiosis**.

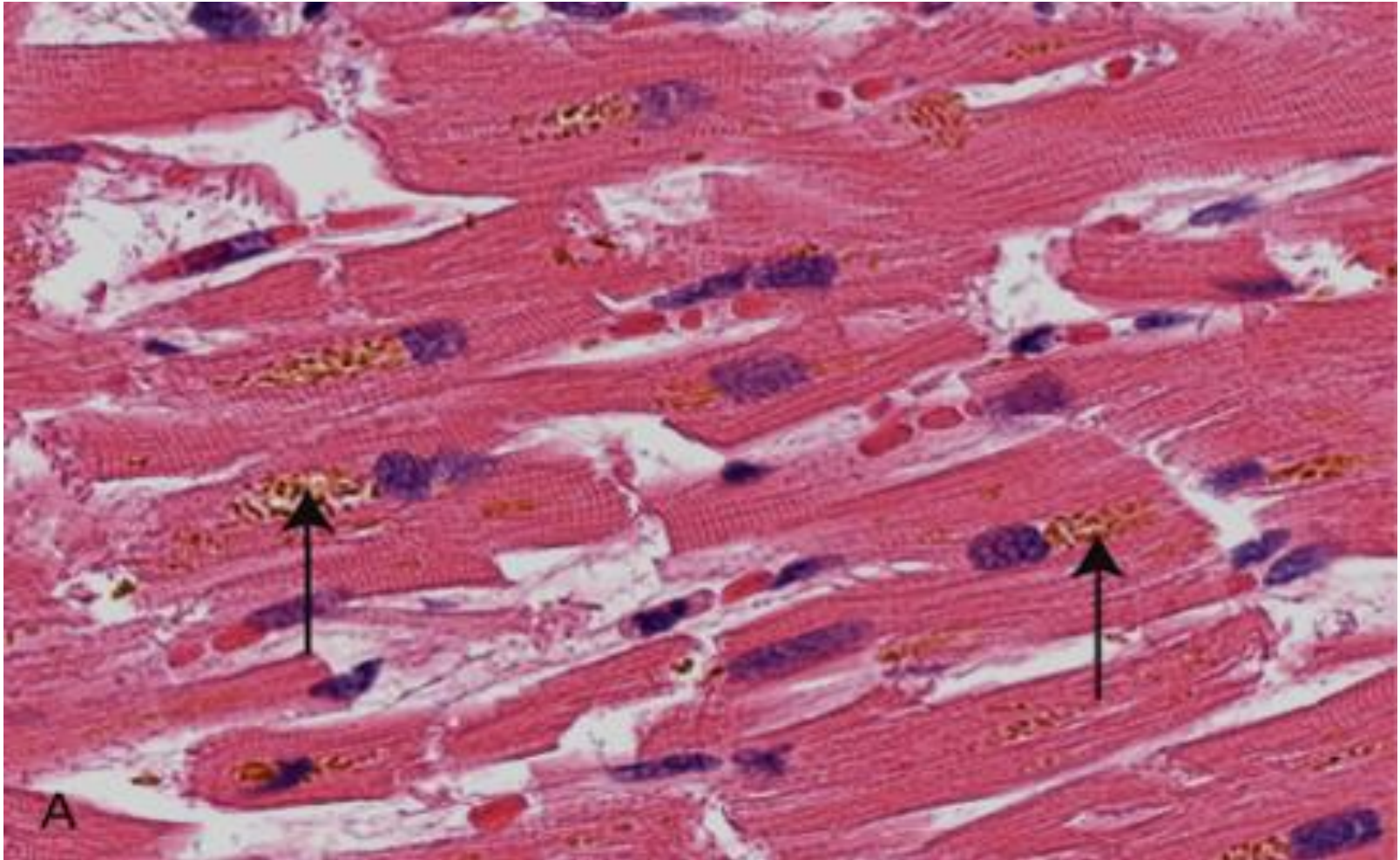
II. Endogenous pigments:

1. Lipofuscin.
2. Melanin.
3. Derivatives of hemoglobin.

○ Lipofuscin:

- It is an **insoluble** brownish-yellow **granular** intracellular material.
- Accumulates in a variety of tissues (particularly the heart, liver, and brain) as a **function of age** or **atrophy**.
- Lipofuscin represents complexes of **lipid** and **protein**.
- It is **not injurious** to the cell.
- is important as a **marker of past free-radical injury**.

Lipofuscin granules in a cardiac myocytes



Brownish-yellow granular intracellular material (deposits indicated by arrows).

○ Bilirubin

- This is a **normal** major pigment of **bile**.
- is derived from **Hb** (but unlike hemosiderin contains no iron).
- **Jaundice** results from **excess bilirubin** pigment that is distributed throughout all tissues and body fluids.
- In the **liver**, particularly when there is **obstruction** to the **bile flow** (e.g. obstruction of the common bile duct):
 - bilirubin is seen within:
 1. Bile canaliculi,
 2. Kupffer cells
 3. Hepatocytes as green-brown globular deposits.
 - This imparts **greenish color** to the liver **grossly**.

○ Melanin

- It is an endogenous, brown-black pigment.
- Produced in melanocytes following the oxidation of tyrosine to dihydroxyphenylalanine.

○ Hemosiderin

- This **iron-containing pigment** consists of aggregates of **ferritin**.
- It appears in tissues as **golden brown** amorphous aggregates.
- Identified by its staining reaction (blue color) with **Prussian blue dye**.
- It exists **normally** in small amounts as **physiologic iron stores** within **tissue macrophages** of the bone marrow, liver, and spleen.
- It accumulates **pathologically** in tissues in excess amounts (sometimes massive)

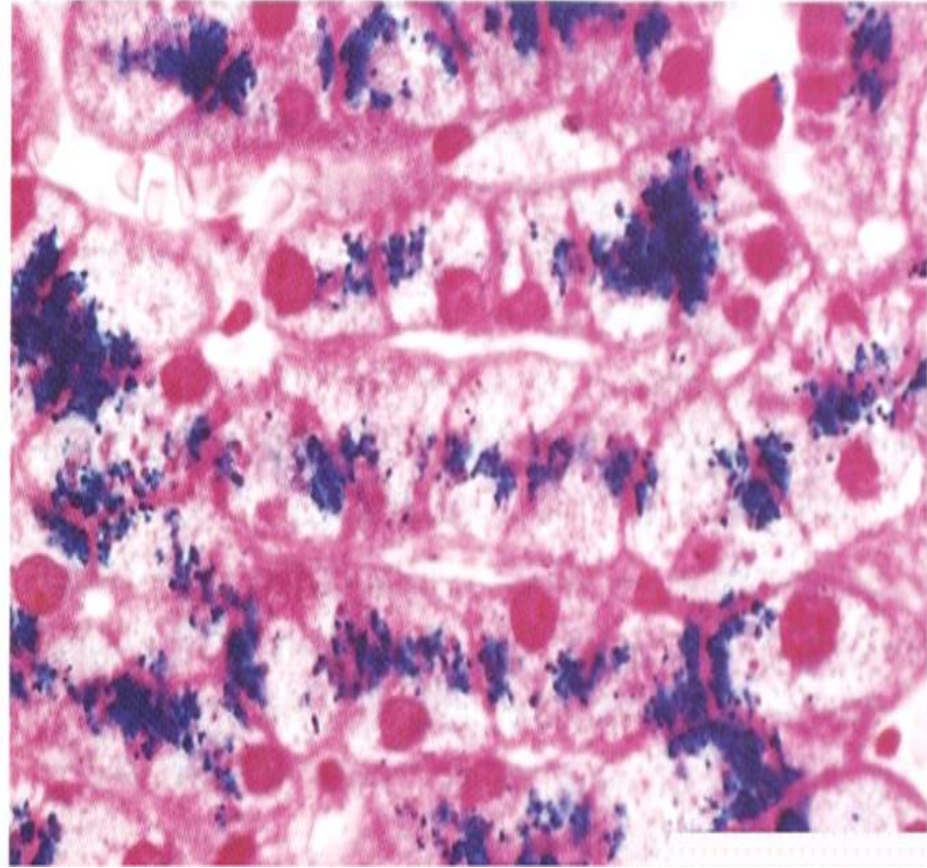
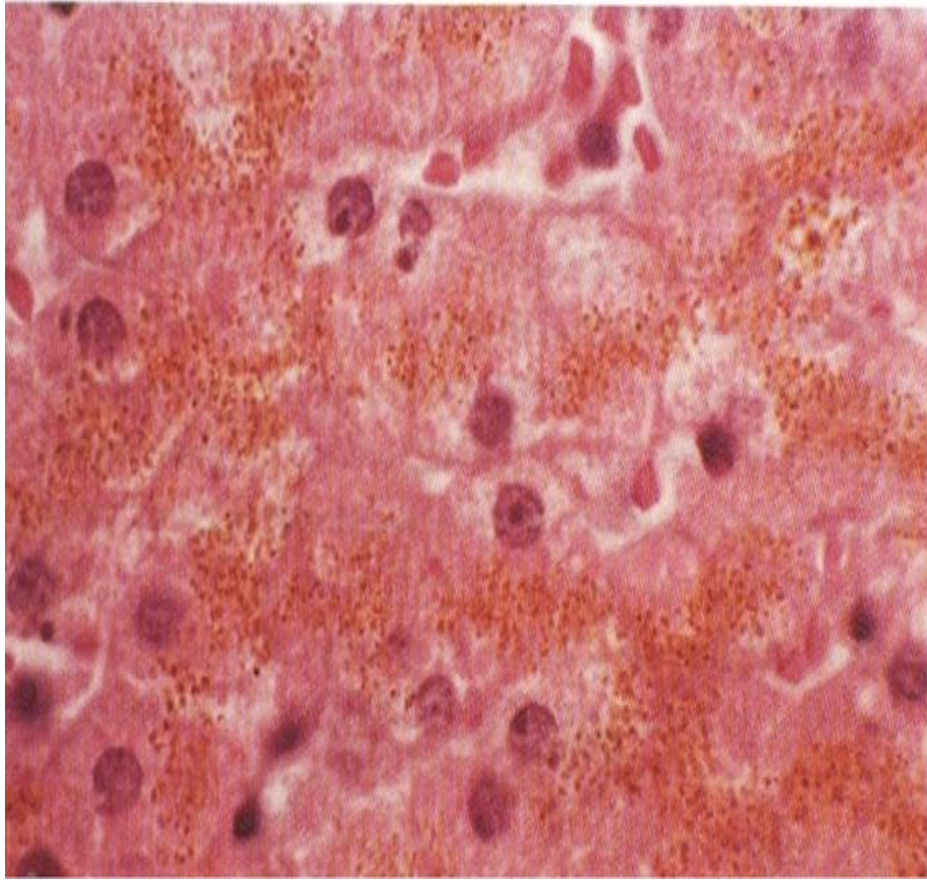
Hemosiderosis:

- Is defined by accumulation of hemosiderin, primarily within **tissue macrophages**.
- **Not** associated tissue or organ damage.

Hemochromatosis:

- Is more extensive accumulation of hemosiderin, often within parenchymal cells and associated with tissue damage, scarring, and organ dysfunction.
- Could be **hereditary (primary) hemochromatosis**.
- Or **secondary hemochromatosis** in multiple blood transfusions

Hemosiderin granules liver cells



LF: Hematoxylin-eosin-stained section showing golden-brown, finely granular pigment.

RT: Iron deposits shown by a special staining process called the Prussian blue reaction.

Pathologic calcification

- It is **abnormal** deposition of **calcium salts**.
- It is a common process in a wide variety of disease states.

I. Dystrophic calcification:

- When the deposition occurs in **dead** or **dying tissues**.
- It occurs in the **absence** of calcium metabolic derangements (i.e., with normal serum levels of calcium).

II. Metastatic calcification:

- The deposition of calcium salts in **normal tissues**.
- Almost always reflects some derangement in calcium metabolism (hypercalcemia).

Dystrophic Calcification

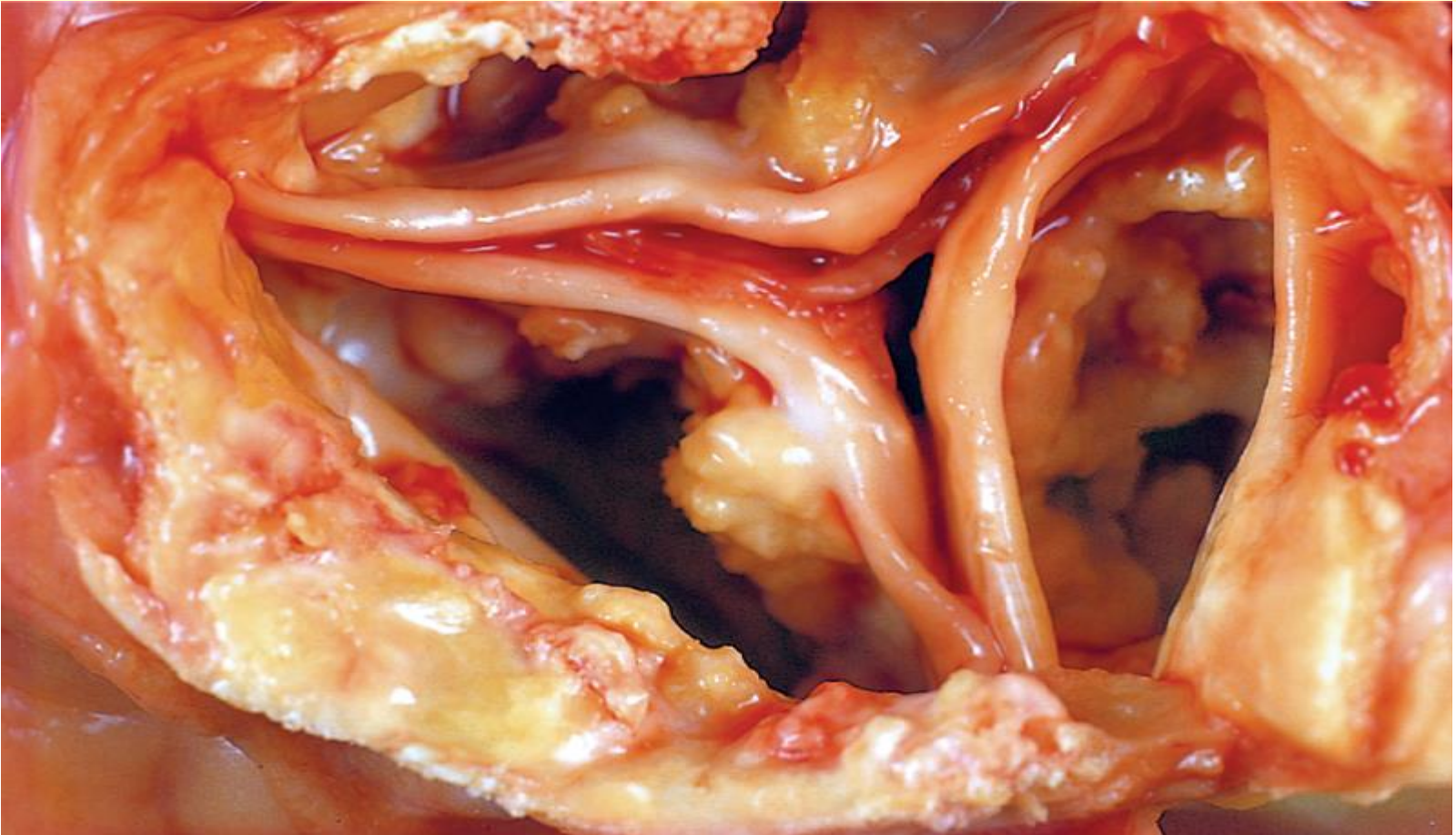
Dystrophic calcification is encountered in:

1. Areas of **necrosis**
 2. Advanced **atherosclerosis**
 3. **Aging** or **damaged heart valves**
- **Dystrophic calcification** of the aortic valves is an important cause of aortic stenosis in the elderly.

Grossly: Fine white granules or clumps, often felt as **gritty deposits**.

Microscopically: appears as intracellular and/or extracellular basophilic **bluish deposits**.

Calcification of the aortic valve.



A view looking down onto the unopened aortic valve in a heart with calcific aortic stenosis.

- The semilunar cusps are thickened and fibrotic.
- Behind each cusp are large, irregular masses of dystrophic calcification that will prevent normal opening of the cusps.

Metastatic Calcification

- The four major causes of hypercalcemia are
 1. Increased secretion of parathyroid hormone
 2. Destruction of bone
 3. Vitamin D-related disorders
 4. Renal failure
- Metastatic calcification can occur widely throughout the body but principally affects the **interstitial tissues** of the vessels, kidneys, lungs, and gastric mucosa.

